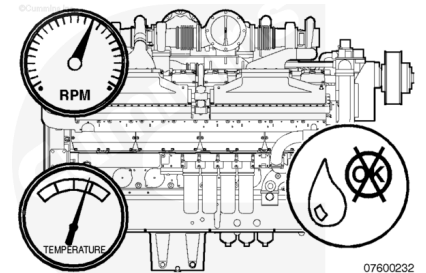


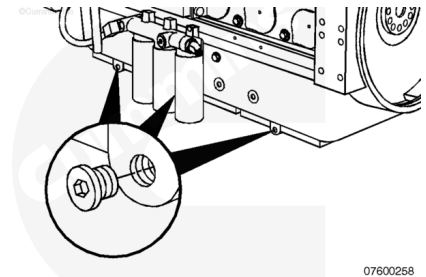
Trainings ▾

Drain

Operate the engine until the water temperature reaches 60°C [140°F].

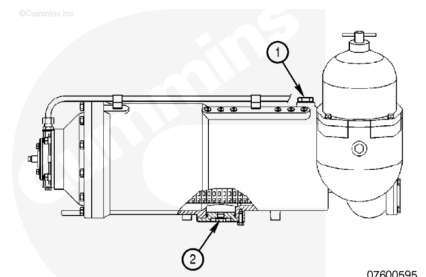


The QSK45 and QSK60 engines have various lubricating oil pan configurations. Remove all drain plugs when draining the oil pan.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Remove the oil drain plugs and drain the oil immediately to make sure all the suspended contaminants are removed from the engine.

For engines with the Eliminator™ filter system, also drain the oil from the Eliminator™ by removing the vent plug (1) and the oil drain plug (2).

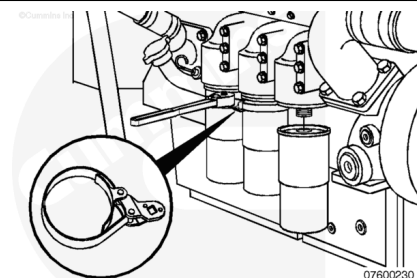
Remove

Note : The Eliminator™ Centrifuge and Bypass Centrifuge do **not** have to be serviced until the 1,000 hour service interval.

The QSK45 engine has three combination oil filters and the QSK60 has four combination oil filters.

Use an oil filter wrench, Part Number 3375049, or equivalent, to remove the oil filters.

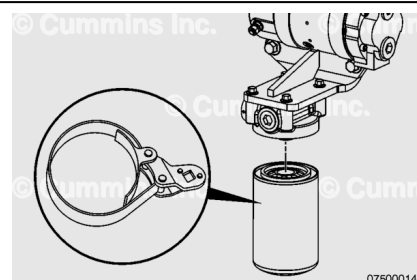
Discard the filter if **not** required for failure analysis.



Some QSK60 Power Generation engines have an oil by-pass line from the PRELUB™ to the fuel pump accessory drive with a small oil filter installed.

Use an oil filter wrench, Part Number 3400158 or equivalent, to remove the oil filter, if applicable.

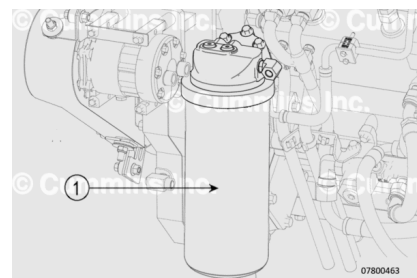
Discard the filter if **not** required for failure analysis.



Some QSK60 engines have a dedicated lubricating oil filter (1) connected to the fuel pump accessory drive.

Use an oil filter wrench, Part Number 3400158 or equivalent, to remove the oil filter, if applicable.

Discard the filter if not required for failure analysis.

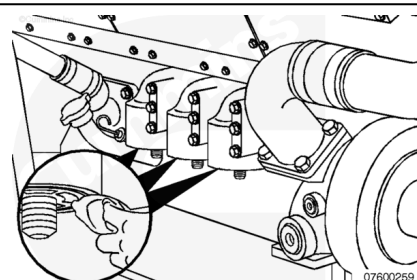


Clean

Note : The Eliminator™ Centrifuge and Bypass Centrifuge do **not** have to be serviced until the 1,000 hour service interval.

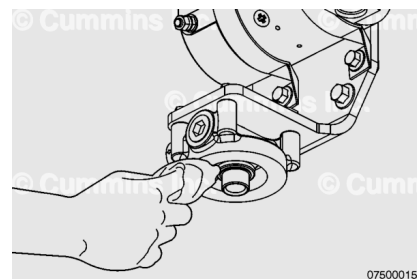
Note : The oil filter o-ring can stick on the filter head. Make sure it is removed.

Clean the oil filter head sealing surface with a lint-free cloth.

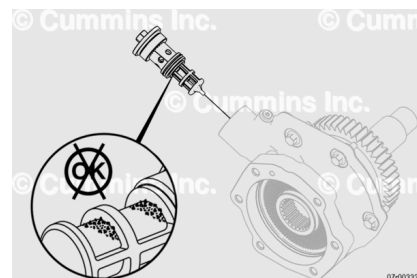


Note : The oil filter o-ring can stick on the filter head. Make sure it is removed.

Clean the oil filter head sealing surface with a lint-free cloth.



Some QSK60 engines have a filter element (1) fitted inside the accessory drive or dedicated lubricating oil filter head. These elements **must** be removed and cleaned with clean engine oil to remove any debris.

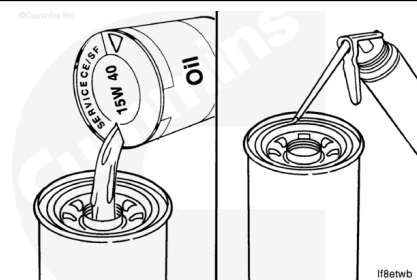


Install

The QSK45 engine has three combination oil filters and the QSK60 engine has four combination oil filters.

Use engine oil to lubricate the oil filter's rubber seal.

Fill the oil filters with clean engine oil.

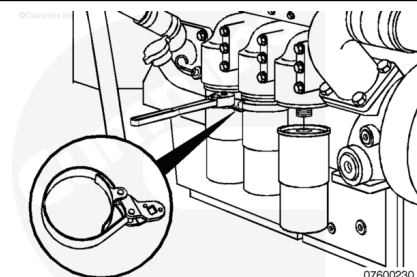


⚠ CAUTION ⚠

Mechanical overtightening can distort the threads or damage the filter element seal. If the threads appear damaged, replace the oil filter head adapter.

⚠ CAUTION ⚠

Locate the oil filter wrench, Part Number 3375049, near the top of the oil filter canister. This will reduce the possibility of damaging the oil filter.



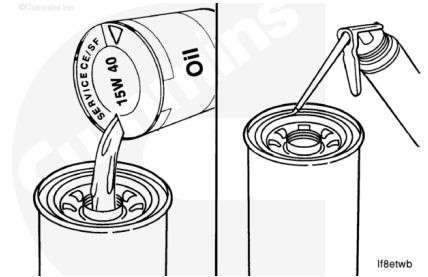
Install the oil filter, turning it until the seal contacts the filter head's sealing surface.

Turn the oil filter an additional 3/4 to 1 full turn.

Some QSK60 Power Generation engines have an oil by-pass line from the PRELUB™ to the fuel pump accessory drive with a small oil filter installed.

Use engine oil to lubricate the oil filter's rubber seal.

Fill the oil filter with clean engine oil.



⚠ CAUTION ⚠

Mechanical overtightening can distort the threads or damage the filter element seal. If the threads appear damaged, replace the oil filter head adapter.



⚠ CAUTION ⚠

Locate the oil filter wrench, Part Number 3400158, near the top of the oil filter canister. This will reduce the possibility of damaging the oil filter.

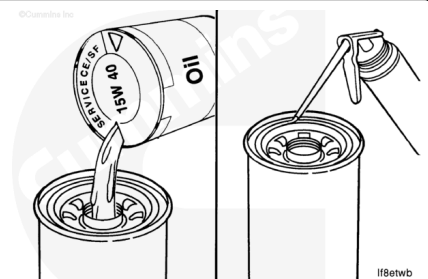
Install the oil filter, turning it until the seal contacts the filter head's sealing surface.

Turn the oil filter an additional 3/4 to 1 full turn.

Some QSK60 Engines have a dedicated lubricating oil filter connected to the fuel pump accessory drive.

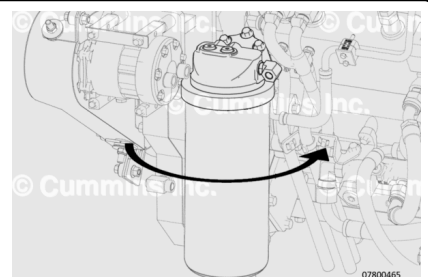
Use engine oil to lubricate the oil filter's rubber seal.

Fill the oil filter with clean engine oil.



Install the oil filter. Turn until seal contacts filter head sealing surface.

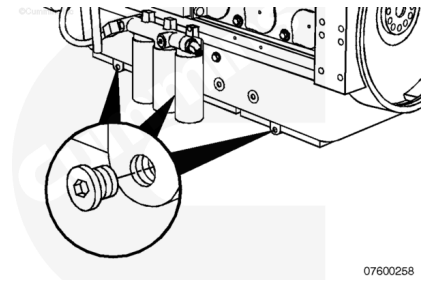
Turn oil filter additional ¾ turn. Use oil filter wrench, Part Number 3400157, or equivalent.



Fill

Install and tighten the drain plug(s).

Torque Value: 47 n•m [35 ft-lb]



Use a high quality 15W-40 multi-viscosity engine oil that meets the American Petroleum Institute (API) performance classification CE or CF4 or better.

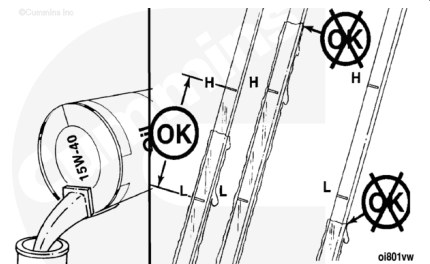
Choose the correct oil for the operating climate. Refer to Procedure 018-017 in Section V.

(/qs3/pubsys2/xml/en/procedures/56/56-018-017.html)



Fill the engine to the correct level with clean engine oil. Refer to Procedure 018-017 in Section V for engine oil capacity. (/qs3/pubsys2/xml/en/procedures/56/56-018-017.html)

Check the oil level on the dipstick. The level **must** be to the high (H) mark on the dipstick. Do **not** overfill. Drain off excess oil in a pan.



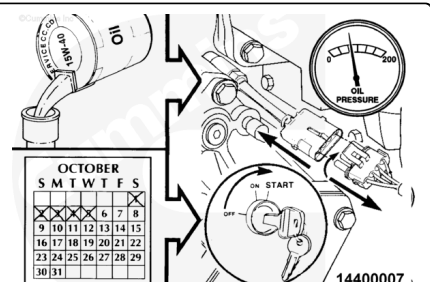
To properly check the oil level:

- Remove the dipstick and wipe clean.
- Insert dipstick fully and remove to observe reading.
- Repeat to make sure of consistent readings.

Note : The o-ring seal on the dipstick can cause a vacuum during fill and falsely create a high reading. Vent the dipstick during fill to prevent a vacuum and false reading.

⚠ CAUTION ⚠

Before starting engines not equipped with a PRELUB™ system, complete the following steps. Engine damage will occur because of inadequate lubrication at startup if these steps are not followed.



Note : This procedure will cause fault codes for the unplugged sensors to be logged. They will go inactive and have no effect on engine operation after the sensors are connected.

Note : Engines that are equipped with a PRELUB™ system are self-priming and do not require completion of this and the next step.

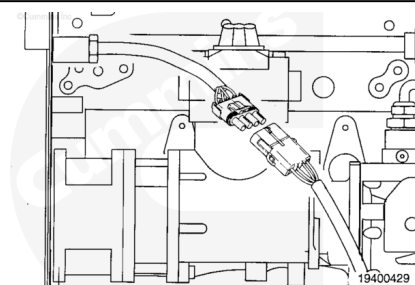
The following steps **must** be completed after each oil change or after the engine has been shut down for more than five days to make sure the engine receives the correct oil flow through the lubricating system.

1. On engines using mechanically actuated injectors, disconnect the engine position sensor.
2. On engines using electronically actuated injectors, the following two sensors need to be disconnected for priming:
 - Engine speed sensor located in the flywheel housing
 - Camshaft position sensor located above the fuel pump on the left bank side.

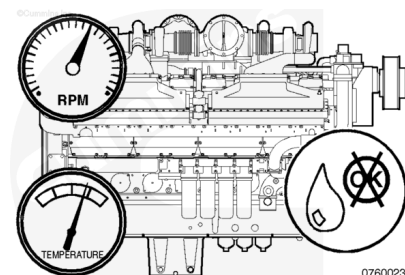
Use the starter motor to rotate the engine crankshaft until oil pressure is indicated on the gauge or the warning light goes out.

Connect the engine position sensor on engines with mechanically actuated injectors.

Connect the engine speed and camshaft position sensors on engines with electronically actuated injectors.



Operate the engine at idle speed to inspect for leaks at the filters and drain plug(s).

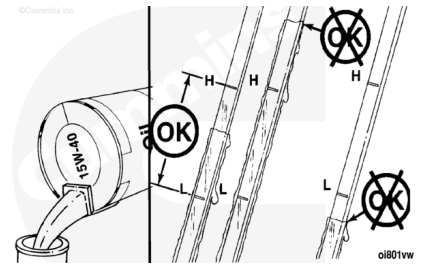


Turn keyswitch OFF.

Wait approximately 5 minutes to allow the oil to drain from the upper parts of the engine.

Check the oil level.

Add oil as necessary to bring the oil level to the high (H) mark on the dipstick.



Last Modified: 04-Jan-2023
